Experiment No. 3

Tutorial Title: Perform Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, and Merge Sort using C
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1. Bubble Sort:

Program:

```
#include<stdio.h>

void bubbleSort(int arr[], int size)
{
   int outer, inner, temp;

   for(outer = 0; outer < size-1; outer++)
   {
      for(inner = 0; inner < size-outer-1; inner++)
      {
         if(arr[inner] > arr[inner+1])
         {
            temp = arr[inner];
            arr[inner] = arr[inner+1];
            arr[inner+1] = temp;
         }
      }
   }
}

int main()
{
   int arr[50], size, idx;

   printf("Enter number of elements: ");
   scanf("%d", &size);

   for(idx = 0; idx < size; idx++)
      scanf("%d", &arr[idx]);

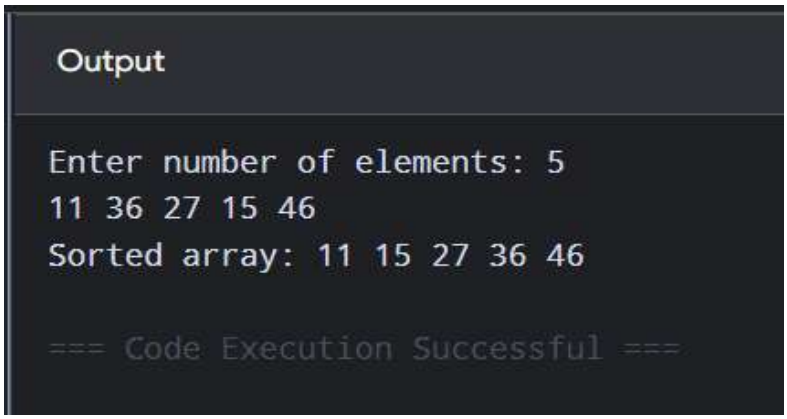
   bubbleSort(arr, size);
```

```
    printf("Sorted array: ");

    for(idx = 0; idx < size; idx++)
        printf("%d ", arr[idx]);

    return 0;
}
```

Output:



2. Selection Sort:

Program:

```
#include<stdio.h>

void selectionSort(int arr[], int size)
{
    int i, j, minPos, temp;

    for(i = 0; i < size-1; i++)
    {
        minPos = i;

        for(j = i+1; j < size; j++)
        {
            if(arr[j] < arr[minPos])
                minPos = j;
        }

        if(minPos != i)
        {
            temp = arr[i];
            arr[i] = arr[minPos];
            arr[minPos] = temp;
```

```
        }
    }
}

int main()
{
    int arr[50], size, i;

    printf("Enter number of elements: ");
    scanf("%d", &size);

    for(i = 0; i < size; i++)
        scanf("%d", &arr[i]);

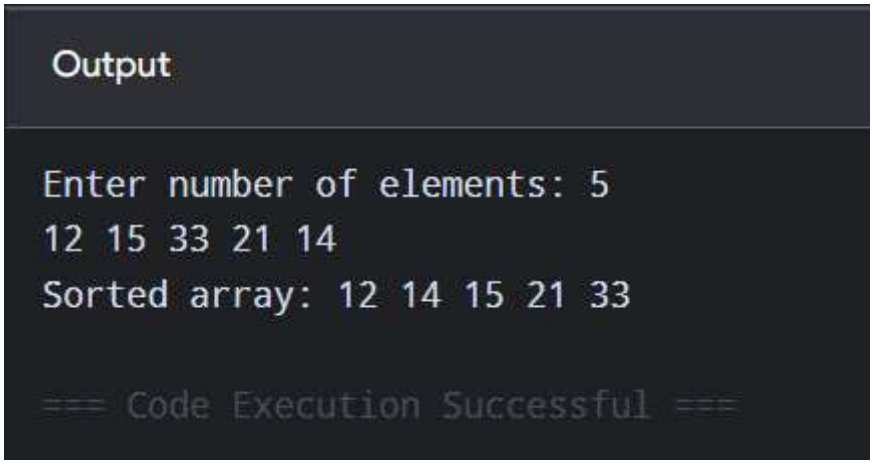
    selectionSort(arr, size);

    printf("Sorted array: ");

    for(i = 0; i < size; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

Output:



3. Insertion Sort:

Program:

```
#include<stdio.h>

void insertionSort(int arr[], int size)
{
   int i, j, key;

   for(i = 1; i < size; i++)
   {
      key = arr[i];
      j = i - 1;

      while(j >= 0 && arr[j] > key)
      {
         arr[j+1] = arr[j];
         j--;
      }

      arr[j+1] = key;
   }
}

int main()
{
   int arr[50], size, i;

   printf("Enter number of elements: ");
   scanf("%d", &size);

   for(i = 0; i < size; i++)
      scanf("%d", &arr[i]);

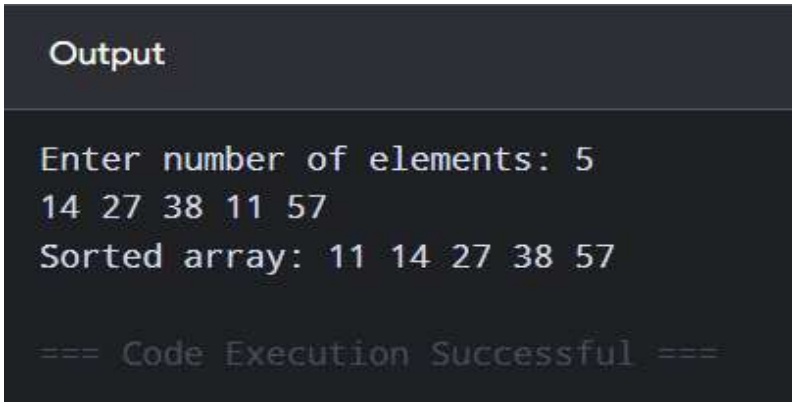
   insertionSort(arr, size);

   printf("Sorted array: ");

   for(i = 0; i < size; i++)
      printf("%d ", arr[i]);

   return 0;
}
```

Output:



4. Quick Sort:

Program:

```
#include<stdio.h>

int partition(int arr[], int low, int high)
{
   int pivot = arr[high];
   int i = low - 1, j, temp;

   for(j = low; j < high; j++)
   {
      if(arr[j] <= pivot)
      {
         i++;

         temp = arr[i];
         arr[i] = arr[j];
         arr[j] = temp;
      }
   }

   temp = arr[i+1];
   arr[i+1] = arr[high];
   arr[high] = temp;

   return i+1;
}

void quickSort(int arr[], int low, int high)
{
```

```
    if(low < high)
    {
        int pi = partition(arr, low, high);

        quickSort(arr, low, pi-1);
        quickSort(arr, pi+1, high);
    }
}

int main()
{
    int arr[50], size, i;

    printf("Enter number of elements: ");
    scanf("%d", &size);

    for(i = 0; i < size; i++)
        scanf("%d", &arr[i]);

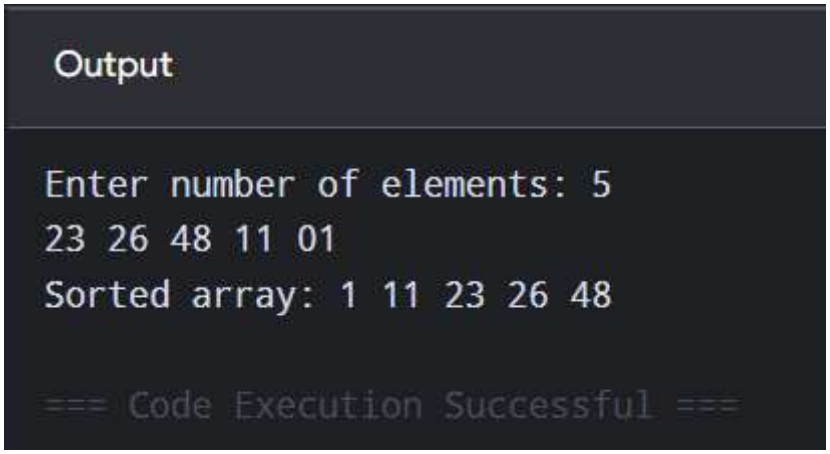
    quickSort(arr, 0, size-1);

    printf("Sorted array: ");

    for(i = 0; i < size; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

Output:



5. Merge Sort:

Program:

```
#include<stdio.h>

void mergeArrays(int arr[], int left, int mid, int right)
{
   int temp[50];
   int i = left, j = mid+1, k = left;

   while(i <= mid && j <= right)
   {
      if(arr[i] <= arr[j])
         temp[k++] = arr[i++];
      else
         temp[k++] = arr[j++];
   }

   while(i <= mid)
      temp[k++] = arr[i++];

   while(j <= right)
      temp[k++] = arr[j++];

   for(i = left; i <= right; i++)
      arr[i] = temp[i];
}

void mergeSort(int arr[], int left, int right)
{
   if(left < right)
   {
      int mid = left + (right-left)/2;

      mergeSort(arr, left, mid);
      mergeSort(arr, mid+1, right);
      mergeArrays(arr, left, mid, right);
   }
}

int main()
{
   int arr[50], size, i;
```

```
    printf("Enter number of elements: ");
    scanf("%d", &size);

    for(i = 0; i < size; i++)
        scanf("%d", &arr[i]);

    mergeSort(arr, 0, size-1);

    printf("Sorted array: ");

    for(i = 0; i < size; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

Output:

